

## RESEARCH RECORD

# QuantLabV2 — Intraday Futures Strategy Research

What we tested, how we tested it, what we found, and what went wrong along the way

NQ and ES 1-minute futures data · discovery 2010-06-07 to 2020-12-31 · unseen testing 2021 to August 2026  
2,659,077 strategy specifications searched · 190,475 frozen · 11,462 survivors · 200 no-edge simulations · 6 finalists

Report date: 22 September 2026 · All figures reproduced from the project's own result files

Official discovery freeze c6da1c8ad6908f9a4ce94d8ed8e4cf5084bc465b94effd750556627be00f1d6b

Final cohort freeze 4e27bc928f39c93d504bcf289ac1ab75a7cdfae5df348d2bdf9c8e4ee7a3c113

**Nothing in this document is a recommendation to trade.** It is a research record. The six strategies at the end are **final forward candidates**: they have not been proven, and their genuinely untested evaluation has not started yet.

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# 1. Executive summary

This project asked one question: **if you search an enormous number of simple intraday trading rules on ten years of history, do any of them keep making money on years they have never seen?** The guiding principle was "*discovery should discover, unseen data should reject*". The system was built so that the newer data physically could not be read until each stage was finished.

|                                       |                                      |                                           |
|---------------------------------------|--------------------------------------|-------------------------------------------|
| STRATEGIES TESTED<br>2,659,077        | FROZEN BEFORE 2021 OPENED<br>190,475 | PROFITABLE IN EVERY UNSEEN YEAR<br>11,462 |
| DISTINCT BEHAVIOURS AMONG THEM<br>259 | NO-EDGE SIMULATIONS RUN<br>200       | FINAL FROZEN CANDIDATES<br>6              |

## What happened, in order

1. **We searched** 2.66 million strategy variants (entry rule + stop + profit management) on NQ and ES 1-minute data from June 2010 to December 2020. The only test was: did it make money after realistic costs? 190,475 unique survivors were frozen, with their rules locked and hashed.
2. **We then opened the later years one at a time** — 2021, then 2022, and so on — running the frozen rules unchanged. After six periods, 11,462 were still profitable in every single year. A coin flip would have left about 2,976.
3. **We checked that against luck.** We generated 200 artificial markets that keep NQ and ES's volatility, calendar, fat tails and even each year's real trend, but destroy any predictable direction. The same 190,475 strategies were run through the identical six-year funnel in every artificial market. The best of those 200 chance worlds produced 4,664 survivors; reality produced 11,462. By the test we had registered in advance, that is **strongly above chance**.
4. **We measured redundancy.** The 11,462 survivors are not 11,462 ideas: they collapse to 259 distinct behaviours, and most of the evidence sits in a few dozen of them.
5. **We selected six finalists** using a rule written and approved before we saw which strategies it would pick, and froze them for a genuine forward test that has not started.

**The honest bottom line.** At the level of the whole population, the result is clearly better than chance, and it is not explained by the rising market: simulations that reproduced each year's real trend produced just as few survivors. But the evidence weakens as you zoom in. Individual survivors look statistically ordinary; the strategies that looked best in the backtest generalised worst; and in 2025 and the partial 2026 the advantage was no longer outside the range of chance. That is why the project ends with a small frozen cohort and an untouched forward period rather than with a conclusion.

## 2. What we set out to test

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### The question

Does a broad, honest search of simple intraday rules on index futures find anything that keeps working on data it has never seen — and can we tell the difference between a real effect and the inevitable luck of testing millions of ideas?

### Why this is hard

- **Testing millions of ideas guarantees winners by luck.** If you test two million coin-flipping strategies, thousands will look brilliant. The best backtest of a large search is usually the luckiest, not the best.
- **It is very easy to cheat accidentally.** Using information that would not have been available at the time — even by one minute — can turn a losing rule into a spectacular one.
- **Costs decide everything intraday.** A strategy that makes 3 points per trade is excellent; one that makes 0.5 points is a loss after commission and slippage.
- **The researcher is part of the experiment.** Someone who has already seen recent markets can unconsciously design searches that suit them.

### The rules we imposed on ourselves, before any result existed

| Rule                                                                               | Why                                                                                                                                         |
|------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| One single test of success: more than zero trades, and positive money after costs. | No profit factor, win-rate, drawdown or "looks nice" filter. Everything else is descriptive only, so the selection cannot be quietly tuned. |
| Keep <b>every</b> profitable candidate, never the best one.                        | Picking the best backtest bakes in the luck. Keeping all of them lets later years do the rejecting.                                         |
| Later years are physically unreadable until their turn.                            | The data loader filters rows inside the file read, so future bars never even reach memory. Every read is logged in an append-only ledger.   |
| Freeze rules and hash them before opening newer data.                              | So nothing can be adjusted afterwards, even unintentionally.                                                                                |
| Write down the test before looking at the answer.                                  | Every stage's pass/fail rule was registered and hashed in advance.                                                                          |

## 3. The system we built

### 3.1 Data

- **Instruments:** E-mini Nasdaq-100 (NQ) and E-mini S&P 500 (ES) futures, continuous front contract.
- **Bars:** 1-minute open/high/low/close. No volume is used anywhere.
- **Period:** 7 June 2010 to August 2026. Files are opened read-only and never modified or copied.
- **Sessions:** a trading day runs 18:00 to 17:00 New York and is labelled by the date it ends on. Daylight saving is handled properly; Sunday evening belongs to Monday's session.
- **Contract rolls:** each bar knows its contract. No calculation, indicator or position ever spans a roll.

### 3.2 How trades are simulated

The simulator is deliberately pessimistic. Every assumption that could flatter a result was set the unfavourable way.

| Assumption                         | Rule used                                                                                                                    |
|------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| Order timing                       | A signal is only known when its bar closes; the order fills at the <b>open of the next minute</b> . Never on the signal bar. |
| Profit targets                     | Fill only if the market trades at least <b>one tick beyond</b> the target. Merely touching it is not a fill.                 |
| Stops                              | Fill at the stop, or worse at the open if the market gaps through it. Never better.                                          |
| Stop and target in the same minute | Which came first is unknowable, so the <b>stop</b> is assumed to have filled.                                                |
| Breakeven stops                    | Move only from the next bar; if the same bar returns to the entry, the trade is assumed stopped at breakeven.                |
| End of day                         | Every position is flat by 16:00 New York. Nothing is ever held overnight.                                                    |

### 3.3 Costs

Costs are charged on every trade, and every strategy is scored under four cost worlds. The official one is BASELINE.

| Cost world      | Commission per round trip | Slippage              | NQ total       | ES total       | Role                                  |
|-----------------|---------------------------|-----------------------|----------------|----------------|---------------------------------------|
| GROSS           | \$0                       | none                  | \$0            | \$0            | diagnostic only                       |
| MODERATE        | \$4.00                    | half a tick each side | \$9.00         | \$16.50        | diagnostic (cost-sensitive watchlist) |
| <b>BASELINE</b> | \$4.00                    | one tick each side    | <b>\$14.00</b> | <b>\$29.00</b> | <b>the official test</b>              |
| STRESS          | \$6.00                    | two ticks each side   | \$26.00        | \$56.00        | diagnostic only                       |

### 3.4 The strategy universe

A candidate is an **entry rule** plus an **initial stop** plus a **profit-management template**. All three were searched together, because the same entry can be profitable or hopeless depending on how the trade is managed.

- **20 entry families** — momentum, mean deviation, breakouts, failed breakouts, fair-value gaps, opening ranges, gaps, candle patterns, prior-session levels, overnight ranges, volatility regimes, multi-timeframe setups, NQ/ES cross-market relationships, time-of-day rules and others. Each has its own parameter grid.
- **Directions:** long only, short only, or both.
- **Stops:** ATR multiples, fixed points, signal-bar, swing, rolling extremes, range fractions, or the family's own level.
- **Management:** 150 templates using a stop (fixed targets from 0.25R to 4R, breakeven rules, five kinds of trailing stop, partial exits, runners, time exits) plus 9 with no stop.
- **Trading windows:** regular hours, morning only, afternoon only, or the full electronic session — always flat by 16:00.

### 3.5 The safeguards

- **Guarded data loader.** The only way strategy code gets prices. It filters rows during the file read, asserts the result is inside the permitted window, makes the arrays read-only, and writes an append-only access log.
- **Permits.** Reading an unseen year requires a permit that only the sequential validator can issue, and only for the next year in sequence.
- **90 automated correctness tests,** covering look-ahead, next-bar entry, the one-tick rule, stop/target ambiguity, breakeven, trailing stops, partial fills, session boundaries, daylight saving, contract rolls, cost arithmetic, profit factor, drawdown, the data guard, deterministic strategy IDs and freeze integrity. An independent second implementation of the trade simulator is used to cross-check the fast one.

## 4. Stage 1 — Discovery (2010-2020)

### How it was run

The search ran in two stages, both on the discovery period only, and both before anything was frozen.

- **Stage A** tested all 5,732 entry specifications with a small standard set of stops and targets: 87,888 variants.
- **Stage B** took every entry that made money *before* costs in Stage A — 5,035 of 5,732 entries — and searched their management far more deeply: 2,571,189 variants. This rule was fixed before Stage A results were seen.

Total search: **2,659,077 specifications**, about 41 hours of compute on a 2-core machine.

### Results

| Quantity                                                                         | Count          |
|----------------------------------------------------------------------------------|----------------|
| Specifications tested                                                            | 2,659,077      |
| Specifications that traded at least once                                         | 2,623,559      |
| Distinct trade behaviours (identical trade lists collapsed)                      | 2,395,009      |
| Profitable before costs                                                          | 905,124        |
| Profitable after MODERATE costs                                                  | 347,069        |
| <b>Profitable after BASELINE costs</b>                                           | <b>229,858</b> |
| <b>Unique candidates frozen</b>                                                  | <b>190,475</b> |
| Profitable even under STRESS costs                                               | 127,127        |
| Cost-sensitive watchlist (moderate but not baseline profitable; never validated) | 110,935        |

So roughly **8.6%** of everything tested made money after realistic costs on the period it was fitted to. That number alone says nothing: it is exactly what a large search over noisy data produces. The frozen cohort split 73.9% NQ / 26.1% ES, and 44% long-only, 32% both-direction, 24% short-only.

The discovery report also recorded the full distributions — profit factor, win rate, target size, trade frequency, drawdown, parameter robustness, entry-versus-management contribution, and breakeven counterfactuals — purely as description. None of it was allowed to influence which candidates were kept.

## 5. Stage 2 — The freeze

Before a single bar of 2021 was read, the 190,475 survivors were frozen. The freeze records, and hashes, everything that could possibly change a result later:

- every strategy's exact entry rule, stop, management template and parameters;
- the execution assumptions (next-bar fills, one-tick rule, pessimistic ambiguity) and the cost configuration;
- the simulator version, the source-code hash, and the hashes of every configuration file;
- the data file fingerprints and the exact discovery date range;
- a SHA-256 over the whole manifest: `c6da1c8ad6908f9a4ce94d8ed8e4cf5084bc465b94effd750556627be00f1d6b`

### Proving the freeze was honest

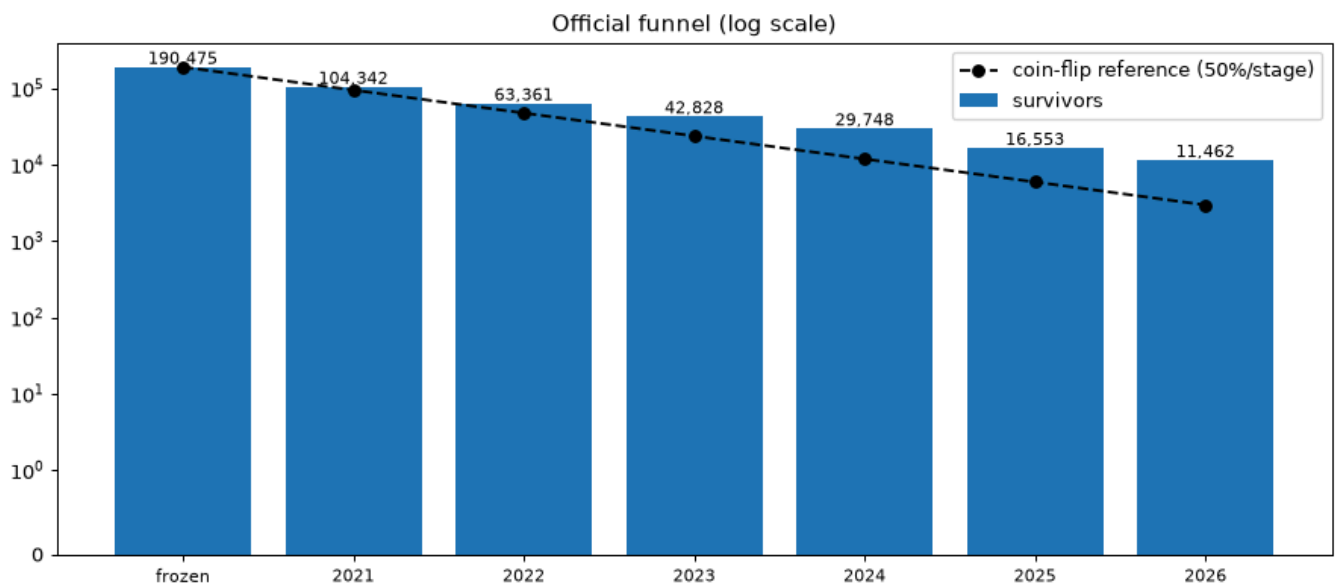
| Check                                                                                   | Result                                                                                                                 |
|-----------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| Independent re-run of a deterministic sample of frozen strategies on the discovery data | <b>601 strategies, 0 mismatches</b> — trades, timestamps, entries, exits, gross and net money, profit factor, win rate |
| Highest real timestamp any research code had read before 2021 was opened                | <b>31 December 2020, 16:59 New York</b> — the last bar of the discovery period                                         |
| Audit of every place in the source code that can open a data file                       | All accounted for; two loopholes found and closed (see Section 10)                                                     |
| Test suite before opening 2021                                                          | 82 tests, 0 failures                                                                                                   |

After this point, no strategy rule could change. Every later stage re-verified these hashes, and all of them still match today.

## 6. Stage 3 — Sequential unseen testing (2021 to August 2026)

Each year was opened only after the previous one was finished. The frozen rules ran unchanged; only the survivors of one year entered the next. The test never changed: more than zero trades and positive money after BASELINE costs.

| Stage                      | Entered | Survived      | Failed | Survival | Share of the original 190,475 |
|----------------------------|---------|---------------|--------|----------|-------------------------------|
| Discovery freeze           | —       | 190,475       | —      | —        | 100%                          |
| 2021                       | 190,475 | 104,342       | 86,133 | 54.8%    | 54.8%                         |
| 2022                       | 104,342 | 63,361        | 40,981 | 60.7%    | 33.3%                         |
| 2023                       | 63,361  | 42,828        | 20,533 | 67.6%    | 22.5%                         |
| 2024                       | 42,828  | 29,748        | 13,080 | 69.5%    | 15.6%                         |
| 2025 (final validation)    | 29,748  | 16,553        | 13,195 | 55.6%    | 8.7%                          |
| <b>2026 (partial year)</b> | 16,553  | <b>11,462</b> | 5,091  | 69.2%    | <b>6.0%</b>                   |



The official funnel. The dashed line is what would remain if every strategy had a 50/50 chance each year (2,976 after six periods).

### What the survivors looked like

- **11,462 survivors** from **567 distinct entry rules**. 94.9% trade NQ.
- 5,577 long-only, 1,227 short-only, 4,658 both directions.
- Median survivor: 574 trades over the six periods, profit factor 1.32, win rate 48%.
- 71% of the survivors' combined money came from long trades.

### Three uncomfortable findings we recorded immediately

1. **Backtest quality did not predict the future.** The rank correlation between a strategy's discovery profit factor and its profit factor in the next unseen year ranged from -0.19 to +0.32. Strategies with a discovery profit factor above 3 survived all six periods only **0.05%** of the time, against **8%** for those below 1.10. The best-looking backtests were the worst performers.

2. **Costs became easier.** As NQ's price level rose, the same rules moved more dollars per trade while commission and slippage stayed fixed. Median net money per trade rose from about \$10 in discovery to \$22-\$121 in the unseen years.
3. **Direction mattered enormously.** Long-biased strategies passed 2021 at 62% versus 40% for short-only — in a period when NQ rose strongly.

Those three facts are exactly why the next stage was necessary. A funnel that ends with 11,462 survivors "beating" a coin flip proves nothing, because the strategies are not independent coins: thousands of them are variants of the same idea, riding the same market.

## 7. Stage 4 — The null funnel: what does pure luck produce?

### The idea

We built artificial markets that look like NQ and ES in every respect except the one thing a strategy needs — predictable direction — and ran **the identical 190,475 frozen strategies through the identical six-year funnel** in each of them. Whatever survives there is what the search procedure produces from nothing.

### How the fake markets were made

Each surrogate minute copies the shape of a real minute from the *same minute of the trading day*, taken from a different day within a three-week window, for NQ and ES simultaneously. What is preserved and what is destroyed:

| Preserved                                                                                                                                                                                                                                                | Destroyed                                                                                                                                    |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| Volatility level and its time-of-day pattern; fat tails; candle shapes; overnight gaps; the exact calendar, session structure, holidays, missing bars and the partial 2026; NQ/ES correlation; and — in half the worlds — each year's exact real return. | The order of market states: any tendency of one move to be followed by another, any time-of-day directional bias, and any up/down asymmetry. |

Two generator families were used (single-minute resampling, and blocks of 5 or 30 consecutive minutes that retain short-horizon behaviour), each in a zero-drift and a trend-preserving version: **200 worlds in total**, all with stored seeds and fully reproducible.

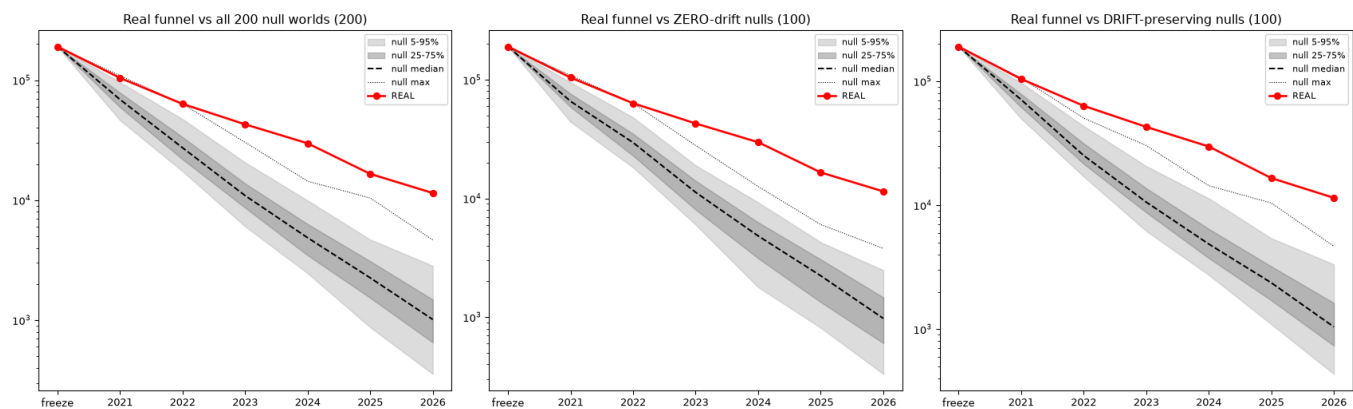
### Proving the fake markets were realistic

Before running the funnel, 144 comparisons of real versus surrogate statistics were made. Volatility, the intraday volatility curve, gaps, ranges, extreme tails and NQ/ES correlation all matched within about 1-9%. Directional structure was gone: rank autocorrelation about zero, 30-minute momentum about zero, no time-of-day bias. Two generator defects were found and fixed at this point (Section 10).

**The most important control:** the same null machinery was run on the *real* 2021-2026 data and reproduced the official funnel exactly — every year's survivor counts and every strategy's trades and money. So any difference we see is the data, not the code.

Results: reality versus 200 chance worlds

| Measure                                             | Real   | Chance worlds: median | 95th pct | Best of 200 | p-value |
|-----------------------------------------------------|--------|-----------------------|----------|-------------|---------|
| Survivors after all six periods                     | 11,462 | 1,013                 | 2,837    | 4,664       | 0.005   |
| Distinct entry rules among them                     | 567    | 145                   | 264      | 374         | 0.005   |
| Strategies with $\geq 200$ trades and $PF \geq 1.3$ | 4,362  | 148                   | 696      | 1,114       | 0.005   |
| Entry rules in that stronger subset                 | 138    | 24                    | 54       | 88          | 0.005   |
| Short-only survivors                                | 1,227  | 150                   | 744      | 1,141       | 0.005   |



The real funnel (red) against all 200 chance worlds, and separately against the zero-drift and the trend-preserving halves. From 2023 onward reality stays above even the best chance world at each stage.

$p = 0.005$  is the smallest value 200 simulations can produce: **reality beat every single one of them on every headline measure**. Under the classification we registered in advance, the result is **STRONGLY ABOVE NULL**.

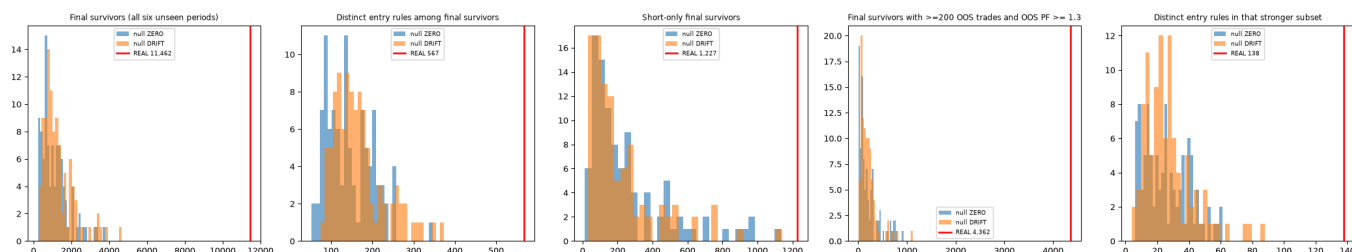
What this ruled out

- **The bull market is not the explanation.** Worlds that reproduced each year's real return exactly produced about as few survivors (median 1,043) as zero-drift worlds (974).
- **Multiple testing and strategy overlap are not the explanation.** Every chance world ran the same 190,475 overlapping strategies through the same six-stage funnel; that is where its ~1,000 survivors come from.

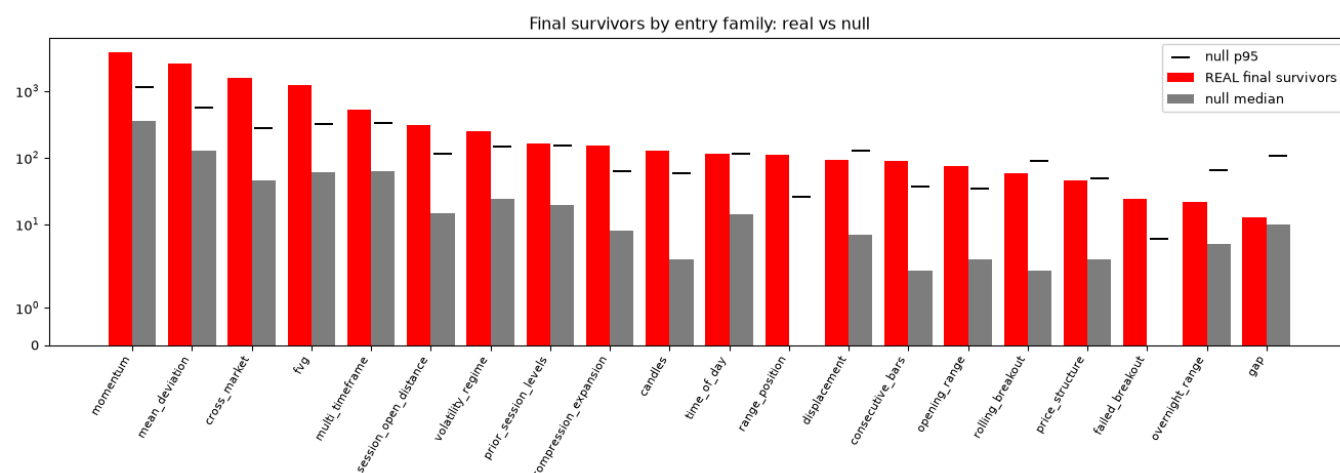
## Where the result is weaker than the headline

The same analysis produced several findings that cut the other way, and they are as important as the headline.

| Finding                                   | Detail                                                                                                                                                                                                                                                  |
|-------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Individual survivors look ordinary</b> | The median real survivor's profit factor (1.32) and win rate (48%) are indistinguishable from those of chance survivors (1.31-1.33, 47-48%; $p \approx 0.4-0.5$ ). A chance world produces ~1,000 survivors that look just like the real ones on paper. |
| <b>Short-only is barely unusual</b>       | 1,227 versus a best chance world of 1,141 — only 8% above. Their entry rules (70) and their median profit factor (1.55) are both inside the chance range.                                                                                               |
| <b>Recent years are not unusual</b>       | Survival rates in 2025 (55.6%) and partial 2026 (69.2%) sit inside the chance range. The advantage was earned in 2022-2024.                                                                                                                             |
| <b>Discovery "stars" are worthless</b>    | Strategies with the highest backtest profit factors or win rates survived at chance rates.                                                                                                                                                              |
| <b>Concentration is normal</b>            | The 95% NQ share and the 71% long share of profits are typical of chance worlds too, so neither is evidence of anything.                                                                                                                                |



Distribution of each headline measure across the 200 chance worlds (blue = zero drift, orange = trend preserved); the red line is reality.



Survivors by strategy family, real versus chance. Momentum, mean deviation, cross-market and fair-value-gap rules stand out; gaps, overnight ranges, breakouts and time-of-day rules do not.

## 8. Stage 5 — How many genuinely different behaviours survived?

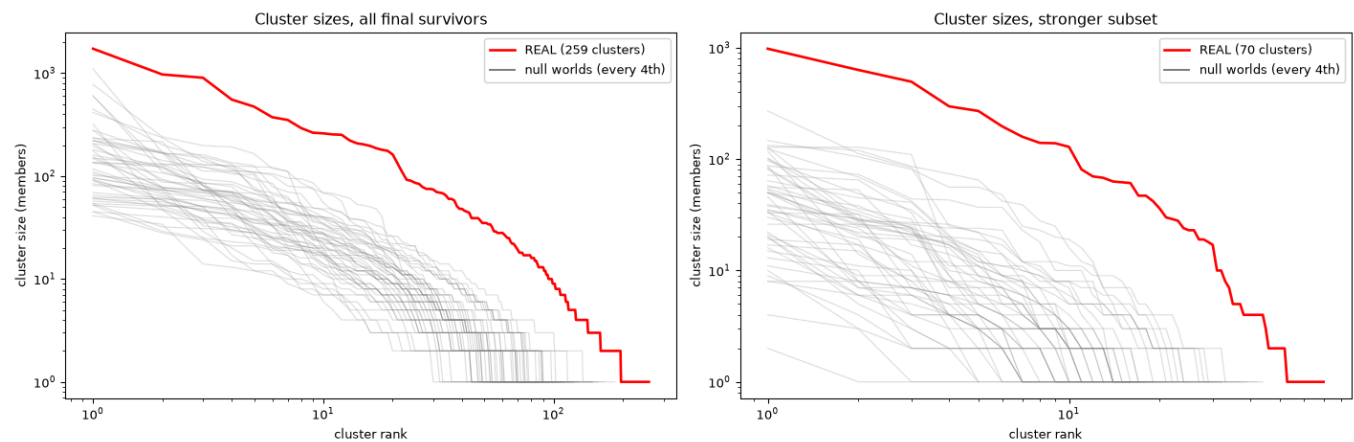
11,462 survivors is a misleading number: hundreds of them are the same idea with slightly different parameters or management. To count real evidence we grouped strategies by **what they actually did** — the correlation of their daily profit and loss over 2021-2026 — not by their labels. Two strategies whose daily results correlate at 0.5 or more were treated as one behaviour. The method, thresholds and pass/fail rule were registered and hashed before any grouping ran.

|                       |                      |                     |
|-----------------------|----------------------|---------------------|
| SURVIVORS             | DISTINCT ENTRY RULES | DISTINCT BEHAVIOURS |
| 11,462                | 567                  | 259                 |
| MEDIAN BEHAVIOUR SIZE | LARGEST BEHAVIOUR    | ONE-OFF SURVIVORS   |
| 4                     | 1,728                | 64                  |

The count is not an artefact of one threshold: at a stricter definition (0.9) there are 2,482 groups, at 0.7 there are 731, at 0.5 there are 259 and at 0.3 there are 94. Grouping by entry-rule overlap instead of profit gives 460 groups, and 78% of those pairs also group together by profit.

### Reality versus chance, at the level of behaviours

| Measure                            | Real | Chance median | Best of 200 | p-value |
|------------------------------------|------|---------------|-------------|---------|
| Distinct behaviours                | 259  | 102           | 184         | 0.005   |
| Behaviours with $\geq 20$ members  | 69   | 12            | 39          | 0.005   |
| Behaviours with $\geq 100$ members | 22   | 1             | 11          | 0.005   |
| High-confidence behaviours         | 38   | 4             | 21          | 0.005   |
| Behaviours in the stronger subset  | 70   | 19            | 48          | 0.005   |



Behaviour sizes in reality (red) against chance worlds (grey), for all survivors and for the stronger subset.

**Two readings, both true.** Reality contains far more distinct surviving behaviours than any chance world — so the excess is spread across dozens of ideas, not concentrated in one lucky family. But when each behaviour is judged on its own against the best behaviour any chance world produced, **none of the 259 is individually beyond chance** (7 are borderline). Only inside the stronger subset — strategies with at least 200 trades and a profit factor of at least 1.3 — do **3 behaviours clearly exceed every chance world**, with 3 more borderline.

### Which kinds of rule carry the evidence

- **Momentum, mean deviation and cross-market (NQ versus ES) rules** beat chance on every measure.
- **Fair-value-gap rules** produce many survivors but almost none with a meaningful sample and profit factor.
- **Gaps, overnight ranges, breakouts, price structure and time-of-day rules** are indistinguishable from chance.
- **High-frequency survivors** (2,500+ trades) form 23 behaviours against a chance 95th percentile of 11 — but 54% of them sit in a single behaviour, and their margins are thin (median profit factor 1.09).
- **Entries, not exits.** Real behaviour groups are more often "one entry idea that works across many different management styles" (76%) than chance groups (64%), and are rarely dependent on one specific management template.

## 9. Stage 6 — The final frozen cohort

The selection rule was written, reviewed and approved **before** it was run, and hashed before it could see which strategies it would choose. In short: take the behaviours that beat chance in the stronger subset; from each, take the most *typical* member (not the most profitable one); discard members whose discovery parameters sat on a knife-edge; and never accept two strategies whose daily results correlate above 0.5.

Applied mechanically, it produced **6 strategies from the 6 eligible behaviours**. No profit, win-rate or recent-performance figure was used at any point.

| # | Strategy         | Behaviour evidence   | Family, direction    | Entry (plain English)                                                                         | Management                                 |
|---|------------------|----------------------|----------------------|-----------------------------------------------------------------------------------------------|--------------------------------------------|
| 1 | C39d55...<br>438 | clearly above chance | mean deviation, long | Buy when the close is stretched at least one volatility unit above its 60-minute average      | Swing-low stop, no target, exit at 16:00   |
| 2 | C175c6...<br>a44 | clearly above chance | momentum, both       | Trade in the direction of a 30-minute move of at least 1.5 volatility units                   | Swing stop; 1.5R trailing stop after +1.5R |
| 3 | C21fa2...<br>1cd | clearly above chance | momentum, both       | Trade in the direction of a sharp 10-minute move of at least 3 volatility units               | Swing stop, target at 1.5R                 |
| 4 | C7162d...<br>6f3 | borderline           | NQ vs ES, both       | Buy NQ when its 30-minute move beats ES's by more than one volatility unit; sell when it lags | Swing stop, target at 2R                   |
| 5 | C3dc5d...<br>3c7 | borderline           | momentum, long       | Buy in the morning when the 4-hour move is at least one volatility unit up                    | Swing stop, target at 2R                   |
| 6 | C1cf9d...<br>41e | borderline           | momentum, both       | Trade in the direction of a 15-minute move of at least 2 volatility units                     | 1 ATR stop; 3 ATR trailing stop from entry |

All six trade one NQ contract, enter only during regular hours (strategy 5 only in the morning), and are always flat by 16:00 New York. Strategy 4 uses ES only as information.

### Historical figures — context only, not evidence

| # | Trades 2021-2026 | Profit factor | Win rate | Max drawdown | Parameter robustness |
|---|------------------|---------------|----------|--------------|----------------------|
| 1 | 381              | 1.70          | 55.1%    | \$18,206     | moderate             |
| 2 | 1,274            | 1.32          | 50.9%    | \$28,987     | broad plateau        |
| 3 | 289              | 1.52          | 51.2%    | \$18,280     | narrow               |
| 4 | 594              | 1.51          | 44.4%    | \$23,342     | moderate             |
| 5 | 293              | 1.38          | 57.0%    | \$15,870     | moderate             |
| 6 | 1,063            | 1.37          | 28.5%    | \$15,277     | broad plateau        |

### Diversification

Maximum correlation between any two finalists is **0.497** (the cap was 0.5), median **0.249**. The cohort is nonetheless concentrated: all six are NQ, intraday, and trade continuation rather than reversal.

### Verification before freezing

- All six were re-run on discovery and on each unseen year: **306 comparisons, 0 mismatches** — trade by trade, including entry and exit timestamps, direction, and gross and net money.

- The cohort manifest is read-only and hashed: 4e27bc928f39c93d504bcf289ac1ab75a7cdfae5df348d2bdf9c8e4ee7a3c113
- The forward runner exists but refuses to run without explicit authorisation, and has not been run.

**The crucial caveat.** Every historical period, including 2021-2026, has now been used to *select* these six strategies. None of it is independent evidence for them any more. Their clean test begins with the trading session of **17 August 2026** (opening Sunday 16 August at 18:00 New York), which no part of this project has ever read.

# 10. Problems we hit, and how we fixed them

This section is deliberately complete, including the mistakes that were caught only by a later check. Two of them would have produced impressive but entirely false results.

## 10.1 Problems that would have corrupted the experiment

| Problem                                                                                                                                                                                                                                                                                      | How it was caught                                                           | Fix                                                                                                                                                             |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Two ways to read data without going through the guard.</b> The <code>doctor</code> command read the whole timestamp column of the data file, and <code>data-check</code> read bars outside the guarded loader. Neither read prices from future years, but both bypassed the safety layer. | The formal source-code audit performed before opening 2021.                 | <code>doctor</code> now reads only file summary information; <code>data-check</code> goes through the guarded loader. Both were disclosed in the audit report.  |
| <b>Identical strategies were not recognised as duplicates.</b> The fingerprint used to detect identical trade behaviour included the exit reason, so two strategies with exactly the same trades looked different.                                                                           | A deduplication test on constructed cases.                                  | The exit reason was removed from the fingerprint; identical trade lists now collapse to one candidate.                                                          |
| <b>A stop-distance list was silently expanded into a full grid,</b> testing many more variants than the configuration described.                                                                                                                                                             | Reading the printed search-size breakdown before launching the real search. | Fixed in the grid expansion, and covered by a regression test.                                                                                                  |
| <b>The surrogate markets' drift was defined slightly wrongly:</b> the trend-preserving worlds missed each year's real return by up to 0.5 percentage points because contract-roll bars were double counted.                                                                                  | The generator diagnostics, run before any surrogate funnel.                 | The drift target was redefined to match the exact return definition; error now below 0.05 points. The superseded diagnostics were archived rather than deleted. |

## 10.2 Results that looked good and turned out to be false

| What we saw                                                                                                                                                               | What it really was                                                                                                                               | What we did                                                                                                                                                                                                  |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| A handful of block-resampled chance worlds showed apparent 30-minute momentum, which would have made the chance worlds unrealistically favourable to momentum strategies. | A few extreme minutes from the April 2025 crash being reused several times by the resampler; on a robust measure the surrogates matched reality. | The diagnostic was switched to a rank-based measure (reported alongside the original), the cause was documented, and it was noted that the bias, if anything, makes the chance worlds <i>harder</i> to beat. |
| The first cluster interpretation stated 3 persistent and 3 recently strong behaviours.                                                                                    | A miscount when reading the table: it is 4 and 2.                                                                                                | Corrected before the report was finalised, and the report regenerated.                                                                                                                                       |
| "2026 data ends on 14 August" for both instruments.                                                                                                                       | True only for ES. The NQ file ends on 10 August, so NQ's 2026 test period is four days shorter.                                                  | Corrected everywhere; every later stage uses per-instrument cutoffs, and the forward start was chosen to be untouched for both.                                                                              |

## 10.3 Bugs in our own checking tools

- **The pre-validation state refused to certify a clean state** because its check searched for the word "failed" and the test summary said "0 failed". Fixed with a proper numeric check.
- **The code audit flagged a legitimate line** that reads an internal results file, not market data. Classified explicitly rather than silenced.

- **The forward-start helper printed the wrong opening day** (Friday instead of Sunday evening) for a Monday session. Fixed; the session date itself was always correct.

## 10.4 Practical and environmental problems

- **Two CPU cores and 8 GB of RAM.** The big searches and the 200 simulations were run as checkpointed background jobs (any crash resumes), memory-mapped data was shared between workers, and the clustering was rewritten to halve peak memory so an  $11,462 \times 11,462$  comparison would fit.
- **Library version differences** (a newer pandas changing timestamp units, a reserved word in the query engine) caused early failures; each was fixed once and covered by tests.
- **A shell command hung silently** waiting for input, so a check script was never created and appeared to "fail". Rewritten and re-run.
- **No version-control identity was configured** on the machine, so the freeze could not record a commit. Rather than invent one, the freeze records source-code and configuration hashes instead.
- **Code changed between discovery and the freeze** (the access ledger was added). Rather than hide it, the freeze records the before and after code hashes and the reason, and the 601-strategy reproduction check proves no trade changed.

## 11. What the evidence supports — and what it does not

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### Supported

- **The survivor population beats chance decisively.** 11,462 survivors against a best-of-200 chance result of 4,664, and the same verdict on every headline measure, including in worlds that reproduce each year's real market trend.
- **The excess is broad, not a single lucky family.** 259 distinct behaviours against a chance best of 184, and 22 large behaviours against a chance best of 11.
- **Three behaviours are individually beyond every chance world** within the stronger subset — two mean-deviation and one momentum, NQ-led — with three more borderline.
- **The effect looks like entry timing, not clever exits.** Real behaviour groups typically work across many different management styles.

### Not supported

- **That any individual strategy has an edge.** On its own statistics, a real survivor is indistinguishable from a lucky one.
- **That the effect is still strong.** 2025 and partial 2026 sit inside the chance range.
- **That short-only, high-frequency or fair-value-gap results are meaningful individually.**
- **That any of this predicts the future,** or that the six finalists will be profitable.

### Limitations we cannot remove

- **The researchers were not blind.** The strategy families were designed by people who had already seen 2021-2026 in an earlier project. No internal test can rule out that this shaped the search space; only genuinely future data can.
- **Execution is simulated.** Fills are modelled at the next bar's open with fixed slippage. Real queues, outages and liquidity shocks are not modelled, though costs were deliberately set pessimistically.
- **The chance worlds are one family of assumptions.** They remove all directional structure and all up/down asymmetry. A strategy exploiting something they remove counts as an "edge" here, even if it is fragile.
- **One market, one era.** Everything is NQ and ES between 2010 and 2026, dominated by a long bull market.

## 12. Where the project stands now

| Stage                              | Status                                                          |
|------------------------------------|-----------------------------------------------------------------|
| Discovery 2010-2020                | <span>COMPLETE</span> 2,659,077 tested, 190,475 frozen          |
| Official freeze                    | <span>COMPLETE</span> hashed, verified, reproduced              |
| Unseen testing 2021 → Aug 2026     | <span>COMPLETE</span> 11,462 survivors                          |
| Archive of the official experiment | <span>COMPLETE</span> read-only, 192 files, hashed              |
| Null funnel (200 worlds)           | <span>COMPLETE</span> strongly above chance                     |
| Edge isolation                     | <span>COMPLETE</span> 259 behaviours, 3 clearly above chance    |
| Final cohort                       | <span>FROZEN</span> 6 strategies, read-only                     |
| Forward test                       | <span>NOT STARTED</span> begins with the 17 August 2026 session |

### What happens next

The only remaining step is to let time pass and run the six frozen strategies on data that nobody — not the code and not the researchers — has seen. The protocol is fixed in advance:

- Nothing about the strategies may change: no parameters, stops, targets, sessions or costs.
- No strategy may be dropped for performing badly, and none may be added. All six are reported for the whole period, winners and losers.
- There are no "kill rules". The forward period is evidence collection, not management.
- The first formal assessment comes after at least six months of forward sessions and at least 100 forward trades per strategy.
- Every forward data load is logged, and trades whose signal came before the start date can never count as forward evidence.

### What would change our minds

- **Towards "this is real"**: the six strategies, or the behaviours behind them, stay profitable after costs on data that never touched the selection — with a sample big enough to matter.
- **Towards "this was selection after all"**: forward results drift to break-even or worse, which is what the 2025-2026 weakening and the null-like quality of individual survivors already hint at.

## Appendix A — Key numbers at a glance

|                                     | Value                                                   |
|-------------------------------------|---------------------------------------------------------|
| Discovery period                    | 7 June 2010 → 31 December 2020                          |
| Unseen periods                      | 2021, 2022, 2023, 2024, 2025, 2026 (partial)            |
| Last data used                      | ES 14 Aug 2026 16:59; NQ 10 Aug 2026 19:59 (New York)   |
| Entry specifications                | 5,732 (1,938 distinct signal configurations)            |
| Total specifications tested         | 2,659,077 (Stage A 87,888 + Stage B 2,571,189)          |
| Frozen candidates                   | 190,475                                                 |
| Survivors of all six unseen periods | 11,462 (567 entry rules)                                |
| Distinct behaviours among survivors | 259                                                     |
| Chance simulations                  | 200 worlds, 6 generator variants                        |
| Final frozen cohort                 | 6 strategies, max pairwise correlation 0.497            |
| Forward start                       | session of 17 August 2026 (opens 16 Aug 18:00 New York) |
| Automated tests                     | 90, all passing                                         |
| Official baseline cost              | NQ \$14.00, ES \$29.00 per round trip                   |

## Appendix B — Where everything lives

| Folder                                          | Contents                                                                                                 |
|-------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| results/main/                                   | The official experiment: discovery results, freeze, each unseen year, reports and the data-access ledger |
| results/archive/OFFICIAL_EXPERIMENT_2026-09-22/ | Permanent read-only archive, including a full plain-English research record                              |
| results/null_funnel/                            | The 200 chance worlds: design, seeds, diagnostics, per-world results, report                             |
| results/edge_isolation/                         | Behaviour grouping: pre-registrations, cluster tables, per-world null comparisons, report                |
| results/final_cohort/                           | The six finalists: selection audit, rule cards, frozen manifest, forward infrastructure                  |
| results/report/                                 | This document                                                                                            |

## Appendix C — Plain-English glossary

| Term                   | Meaning                                                                                                                 |
|------------------------|-------------------------------------------------------------------------------------------------------------------------|
| Profit factor (PF)     | Money made on winning trades ÷ money lost on losing trades. Above 1 is profitable.                                      |
| Win rate (WR)          | Share of trades that made money after costs.                                                                            |
| R                      | One unit of initial risk: the distance from entry to the initial stop. A 2R target aims to make twice what it risks.    |
| ATR                    | Average True Range: a measure of how much the market typically moves; used to size stops and thresholds.                |
| Drawdown               | The largest peak-to-trough fall in cumulative profit.                                                                   |
| Out-of-sample (OOS)    | Data the strategy was not fitted on. Here: 2021 onward — though after the final selection, it counts as selection data. |
| Surrogate / null world | A simulated market with realistic statistics but no predictable direction.                                              |
| p-value (here)         | The share of chance worlds that matched or beat reality. 0.005 means none of the 200 did.                               |
| Behaviour cluster      | A group of strategies whose daily profits move together (correlation $\geq 0.5$ ) — counted as one idea.                |
| Medoid                 | The most typical member of a group: the one most correlated with all the others.                                        |

**Reproducibility.** Every number in this report comes from files in the folders above. Each stage re-verifies the hashes of the previous ones, and all of them still match. The complete chain — from the first search to the frozen finalists — can be re-derived from the archived configuration, source snapshot and recorded seeds.